

David Cai

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EDUCATION

University of Texas at Arlington

Bachelor of Science in Computer Science

Arlington, TX

May 2027

University of California, Irvine

Full Stack Development Certificate

Irvine, CA

Oct 2018 – Apr 2019

EXPERIENCE

Software Engineer Intern, Technology Internship Program

Jun 2026 – Aug 2026

Capital One

Plano, TX

- Built and shipped an end-to-end data processing pipeline in Python that ingests live loan application data via internal APIs and REST endpoints, processing 100+ cases daily and reducing manual context-gathering time from 5 minutes to 2 minutes per case.
- Developed comprehensive API integration patterns and tested third-party service integrations using Bash scripting and Python, documenting retry logic, error handling, and rate-limiting strategies for production deployment.
- Collaborated with data engineering and backend teams to design database schemas and data flow requirements, ensuring data integrity and efficient retrieval for downstream systems and reporting pipelines.
- Spearheaded end-to-end product discovery through user research and live feedback demos, translating user friction points into prioritized technical requirements, system trade-offs, and architecture decisions.
- Accelerated workflow development and established comprehensive system documentation, architecture diagrams, testing strategies, and handoff procedures to transition the prototype into production.

PROJECTS

GraphT | PostgreSQL, React/TypeScript, Vercel

Jul 2026 – Present

- Architected a full-stack skincare review platform using Next.js 16, Neon Postgres, and Vercel infrastructure to aggregate product data and enable user-driven analytics.
- Designed and implemented a PostgreSQL database schema with relational tables for users, products, ingredients, and trial results, supporting complex queries and efficient data retrieval for product comparison and analytics features.
- Developed a web-scraping pipeline using Python to aggregate and normalize 180k+ skincare products and 20k+ ingredients into structured database tables, with data validation and deduplication logic.
- Integrated third-party APIs (Perfect Corp YouCam skin analysis) into the backend to enable real-time data enrichment and synchronization with the Postgres database.
- Designed a privacy-focused data architecture on Vercel Blob and Neon Postgres, handling secure storage and retrieval of user trial photos, routine logs, and analysis records with encryption and access controls.

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Python, Java, HTML/CSS, SQL, Bash

Databases: PostgreSQL, Informix, Microsoft SQL Server

Frameworks & Libraries: React, Express.js, SQLAlchemy, Node.js

CI/CD & DevOps: Jenkins, GitHub Actions, Git/GitHub, GitLab

Developer Tools: VS Code, IntelliJ, Vercel, Google Cloud Platform, RESTful APIs, PostgreSQL, MongoDB

UI Libraries: Redux, TanStack Query, Recharts, Tailwind CSS, shadcn/ui, Mantine UI, Chakra UI

STUDENT ORGANIZATION

Treasurer - Mobi Byte (Web Development Club)

Jan 2025 – Present

University of Texas at Arlington

Arlington, TX

- Responsible for managing club finances, including taxes, budgeting, funding allocations, purchase approvals, and financial audits.
- Active engagement with overall club strategy, operations, and event planning as part of the leadership team.
- Mentoring new members on development practices and tools, fostering an inclusive learning environment.
- Design and lead technical workshops on web development topics to support member growth and engagement.